

USB Image Acquisition and Hash Verification

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Computer and Digital Forensics

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LAB OVERVIEW

This lab focused on the forensic acquisition of a USB storage image and verification of its integrity using cryptographic hashing. The process involved creating a forensic image of the USB device and generating hash values to ensure the acquired evidence remained unaltered and forensically reliable.

OBJECTIVES

- To carry out image acquisition successfully using both windows and Linux a tools to acquire images,
- To carry out Hashing of the images acquired

TOOLS

- USB drive
- FTK imager
- Kali Linux
- Microsoft Word
- Github
- Md5sum and sha256sum

METHODOLOGY

Firstly, we get a manual USB drive and input it into the system. And then we create a Folder named **CIP-B102-Lab4-Evidence**

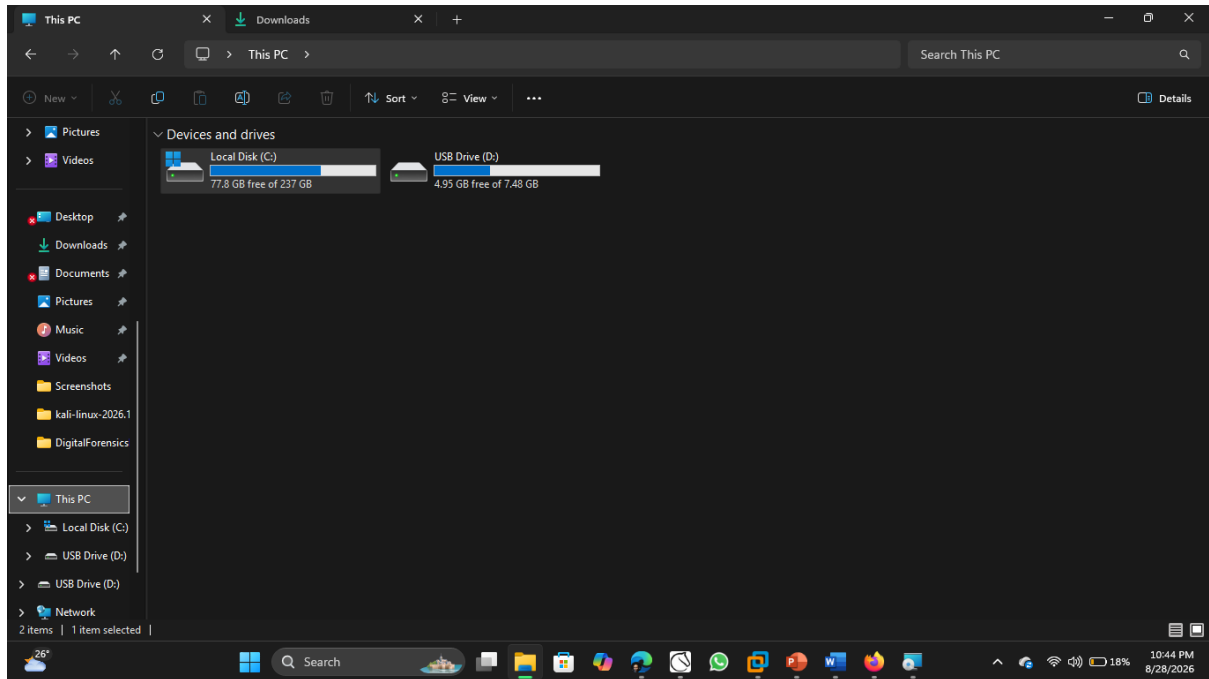


Fig 1.1 USB Drive successfully inserted into the system

Then I Switch to the D: Drive on the command line and I used the dir command to see the list of directories and files on the USB to be able to create the LAB folder appropriately

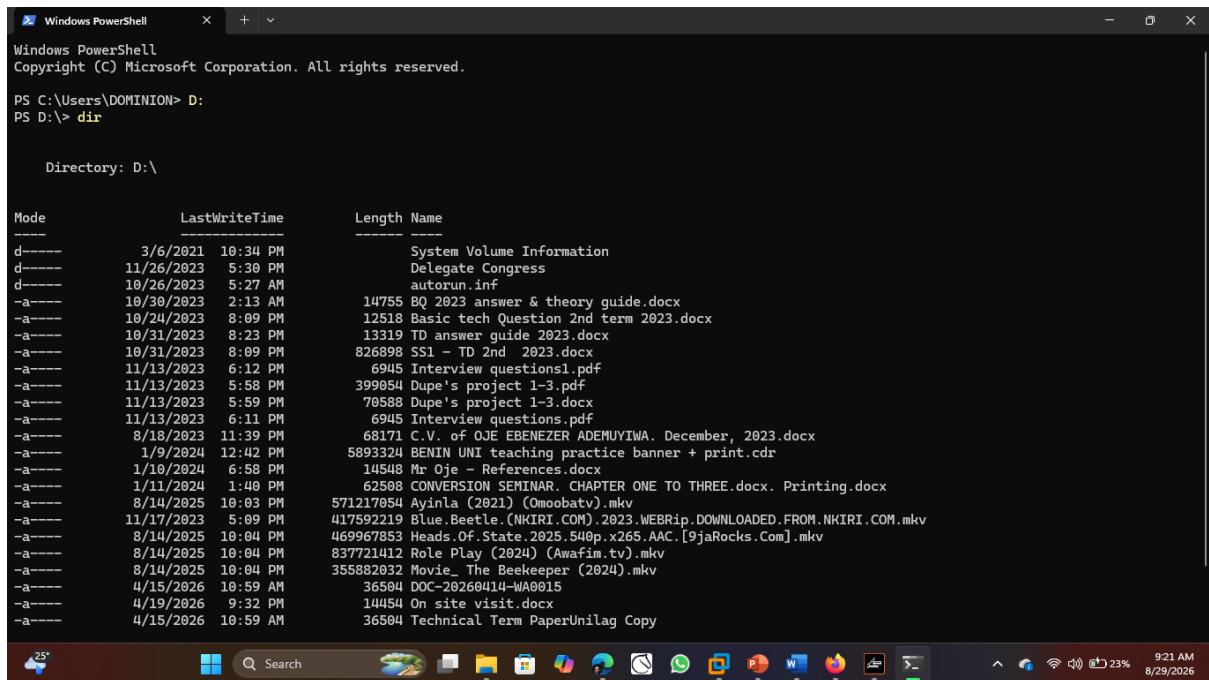
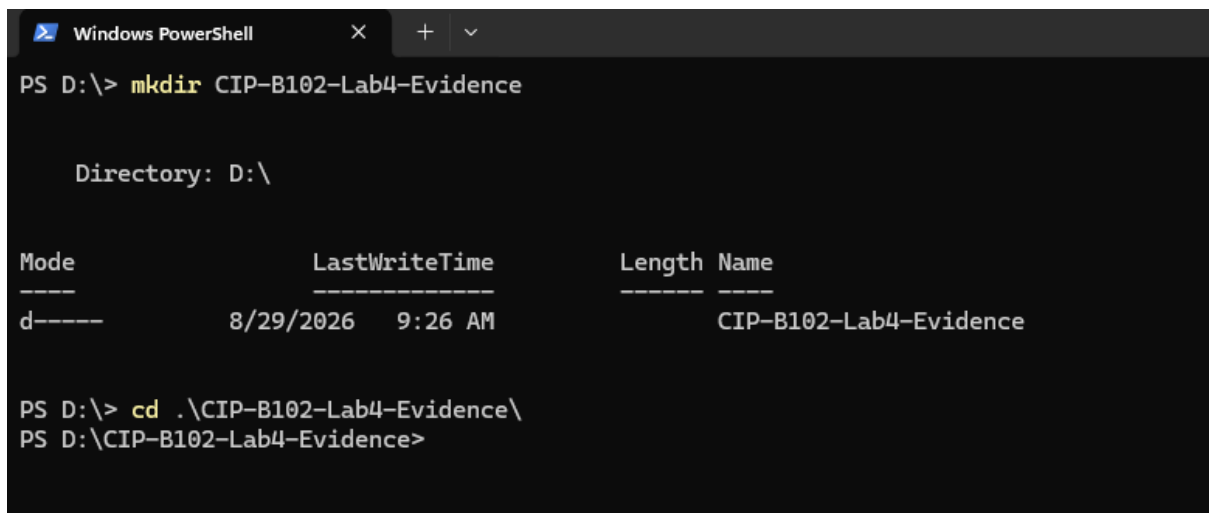


Fig 1.1 Switched to the D: Drive on the Command Line

Then we create the USB Evidence Folder Lab.



```
Windows PowerShell
PS D:\> mkdir CIP-B102-Lab4-Evidence

Directory: D:\

Mode                LastWriteTime         Length Name
----                -
d-----          8/29/2026   9:26 AM                CIP-B102-Lab4-Evidence

PS D:\> cd .\CIP-B102-Lab4-Evidence\
PS D:\CIP-B102-Lab4-Evidence>
```

Fig 1.3 USB Evidence Lab Successfully Created

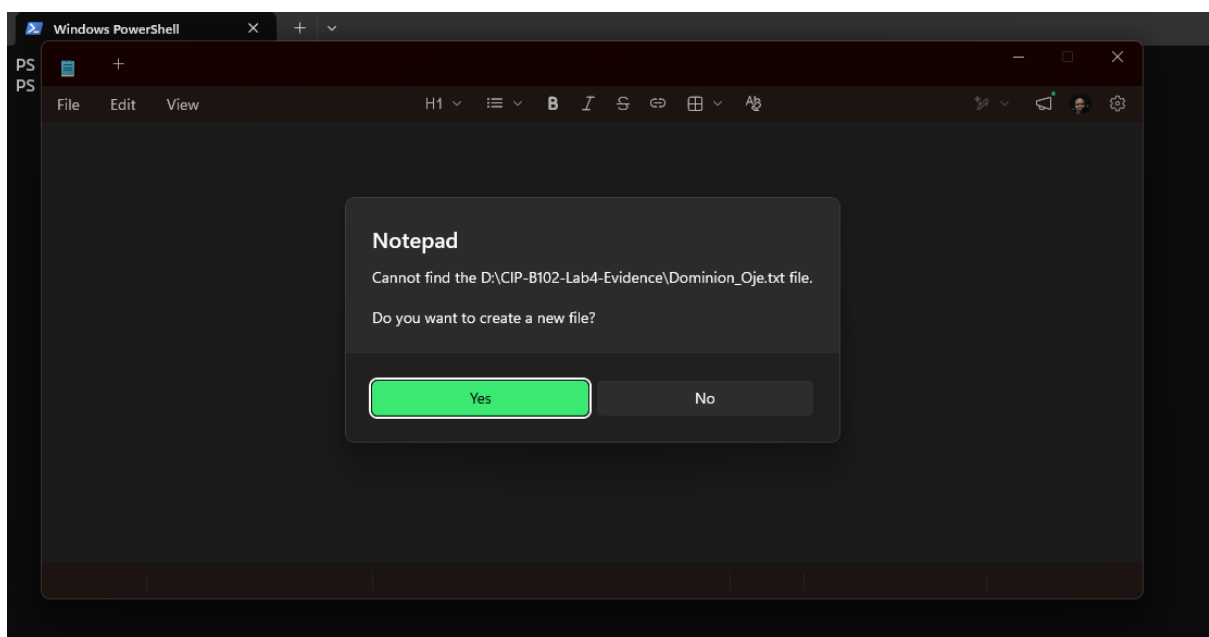


Fig 1.4 Created a File in the Evidence Lab

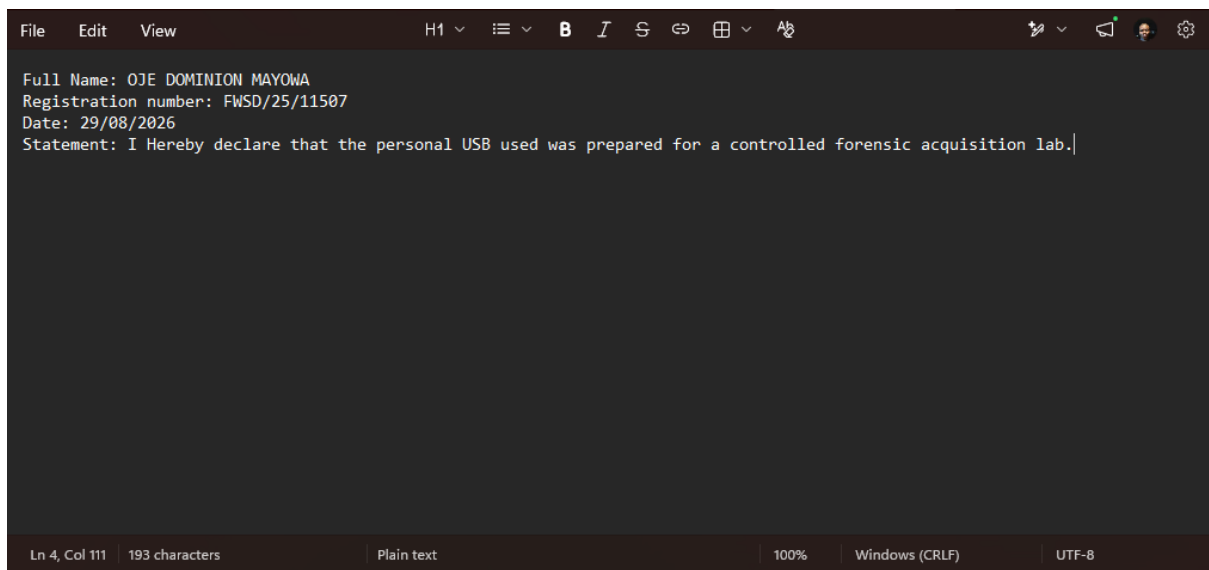


Fig 1.5 File Details

PART B: EVIDENCE DOCUMENTATION

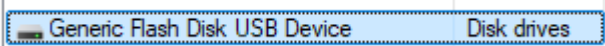
Case/Lab ID	SBT-DF202_LAB2
Student name	OJE DOMINION MAYOWA
USB brand/description	
USB capacity	8 GB Capacity: 8,033,665,024 bytes 7.48 GB
Date and time prepared	Saturday, August 29, 2026 10:01:59 AM
Acquisition method	Bit-by-bit forensic disk imaging
Tool and version	FTK Imager 4.7.3.81
Image filename	USB01.dd

Image storage location	
Hash algorithm used	MD5 and SHA-256

Part C: Acquiring the USB Image

Step 1: Setting up the physical Drive

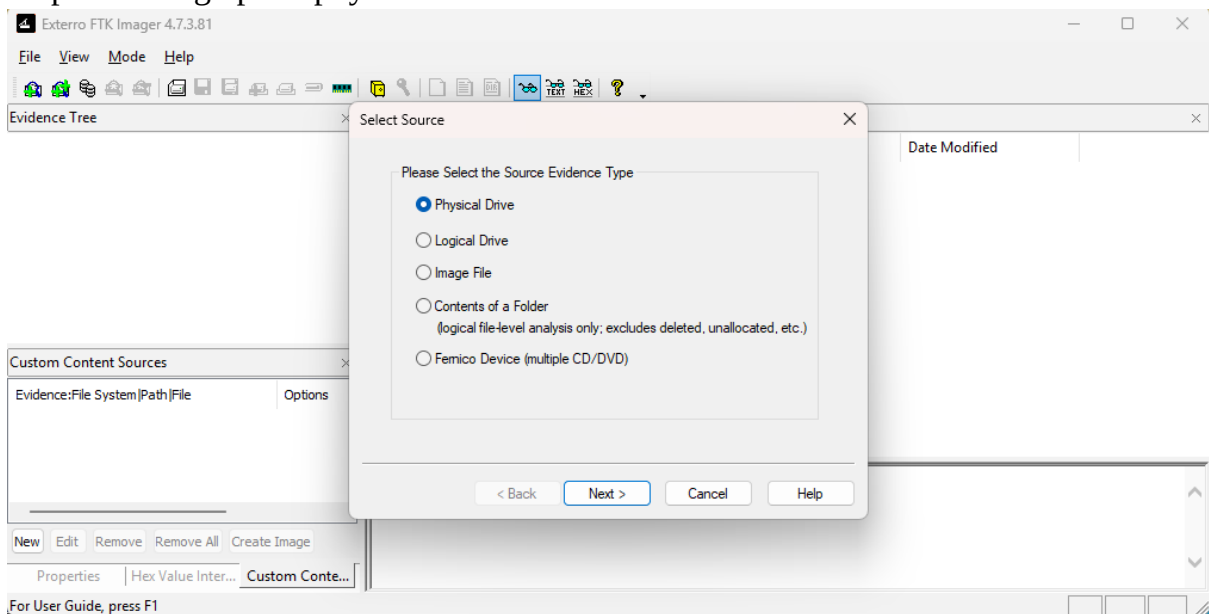


Fig 2.0 Setting up the Physical Drive on FTK Imager

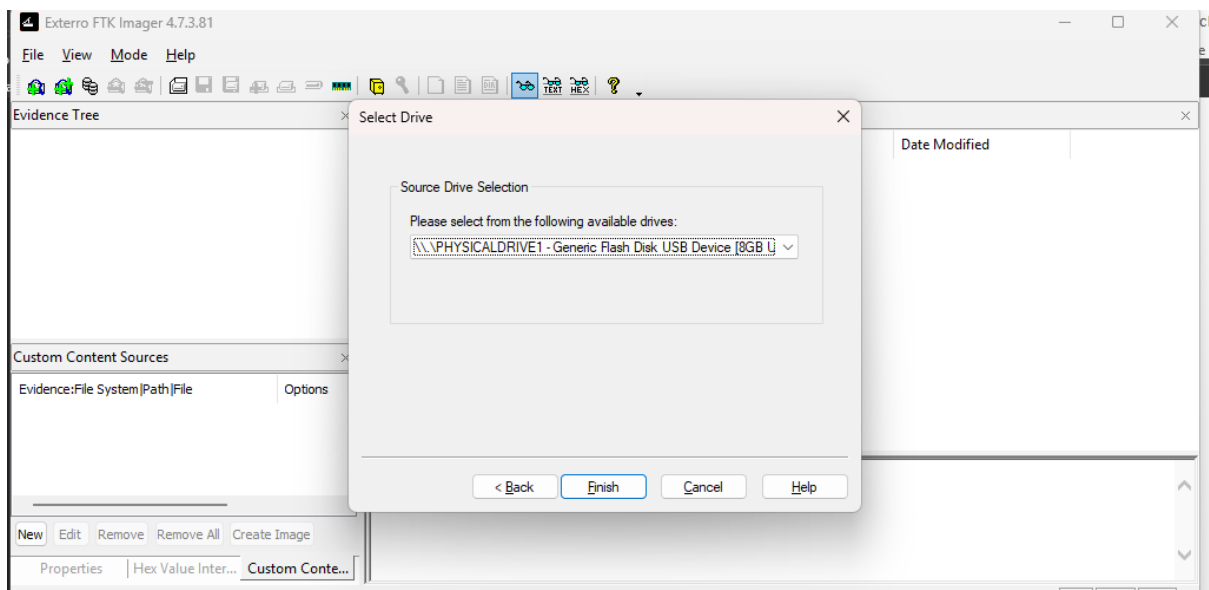


Fig 2.1 Choosing the External USB

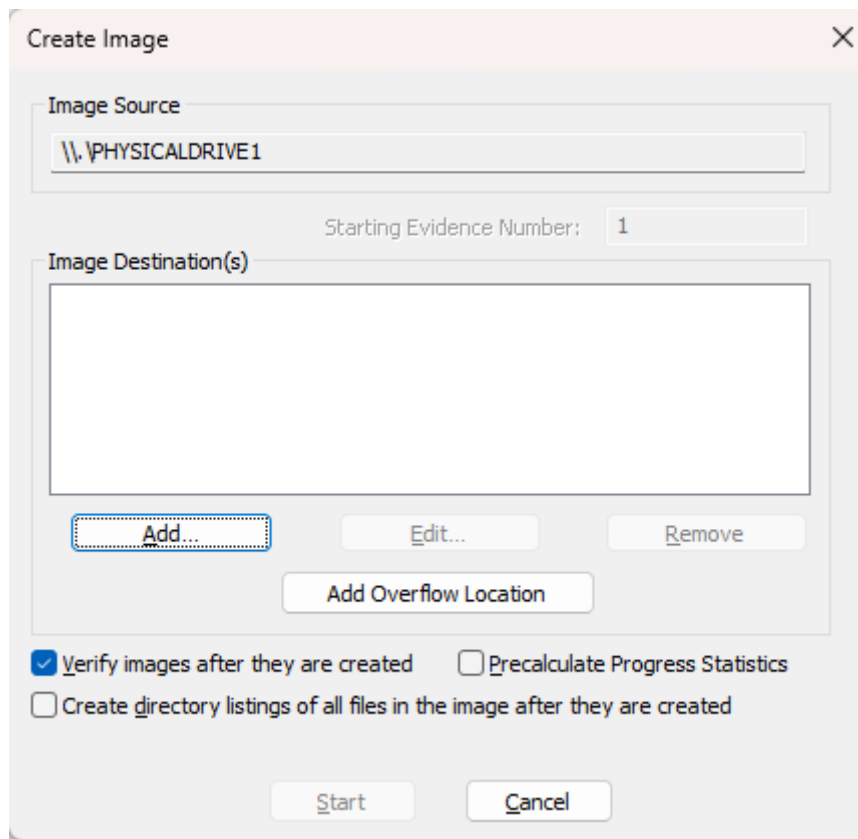


Fig 2.2 Adding an Image Source

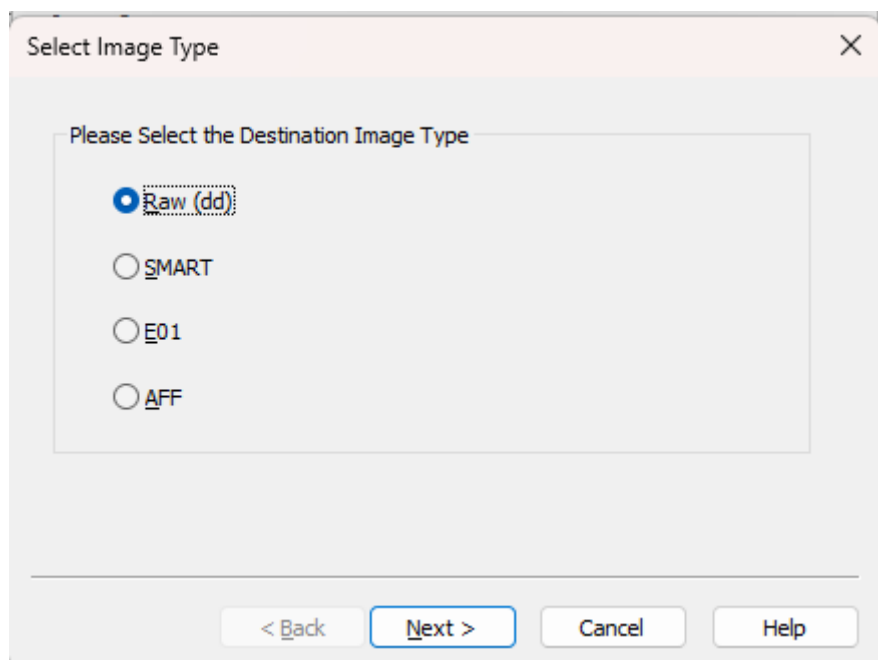
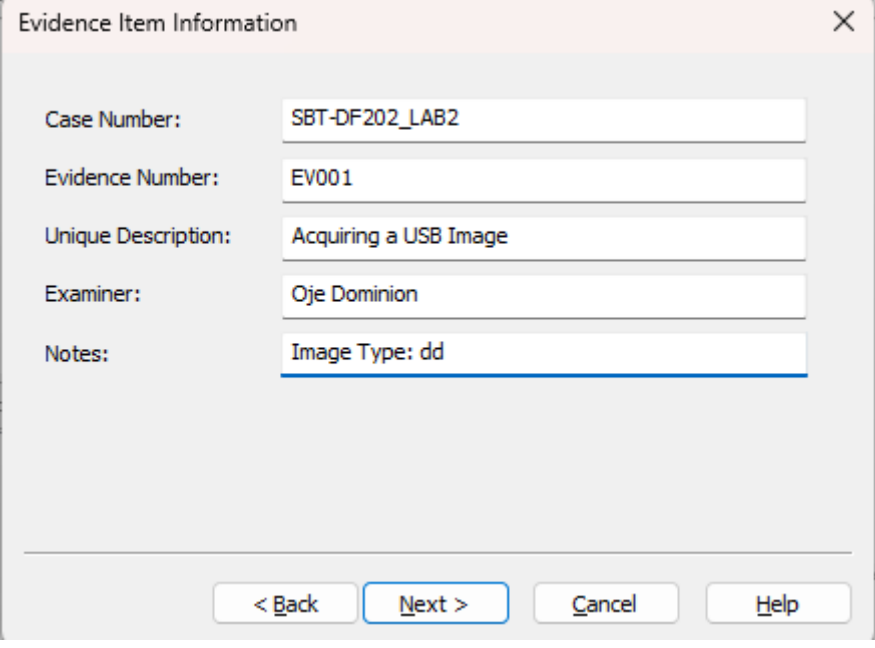


Fig 2.3 Selecting Image Type



The 'Evidence Item Information' dialog box contains the following fields and values:

Field	Value
Case Number:	SBT-DF202_LAB2
Evidence Number:	EV001
Unique Description:	Acquiring a USB Image
Examiner:	Oje Dominion
Notes:	Image Type: dd

Navigation buttons at the bottom: < Back, Next > (highlighted), Cancel, and Help.

Fig 2.4 Evidence Information

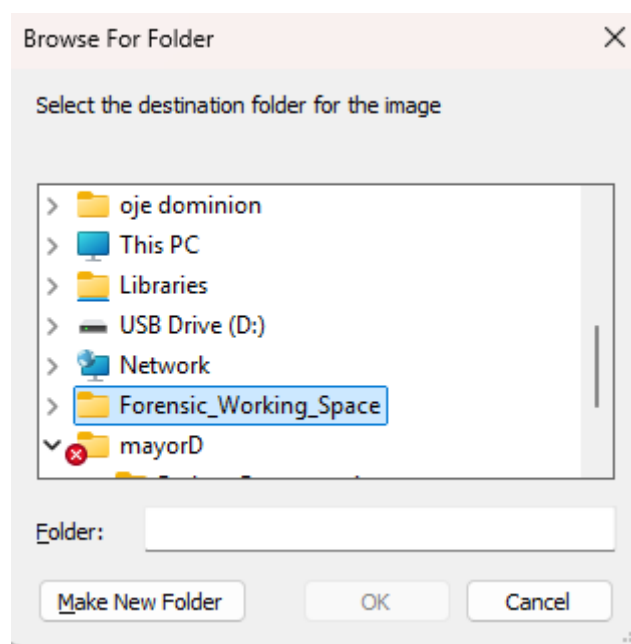


Fig 2.5 Image Destination

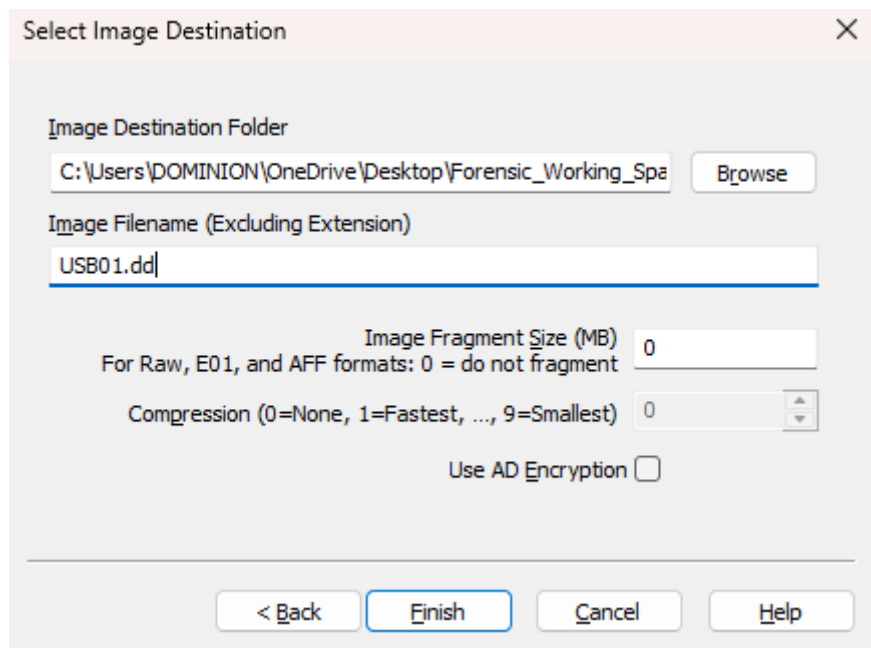


Fig 2.6 Selecting Image Destination and Image File Name

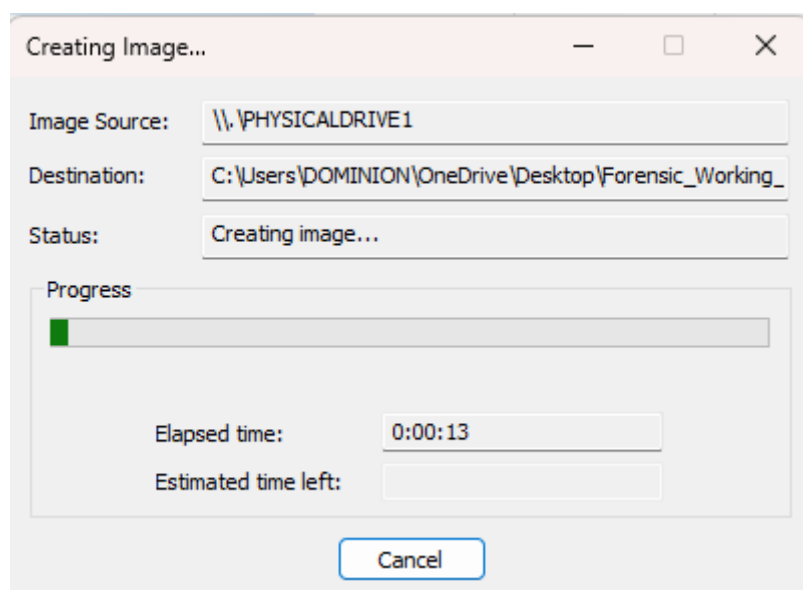


Fig 2.7 Image Creation Process Successfully Begins.

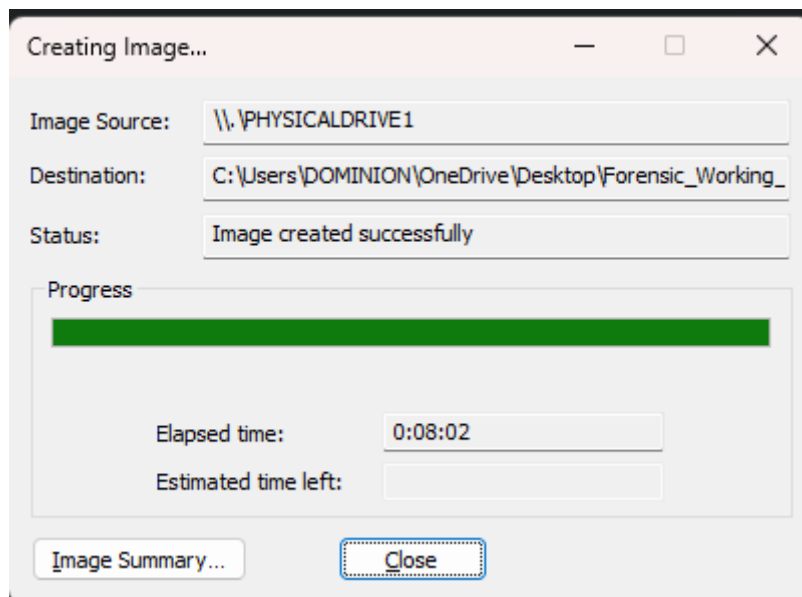
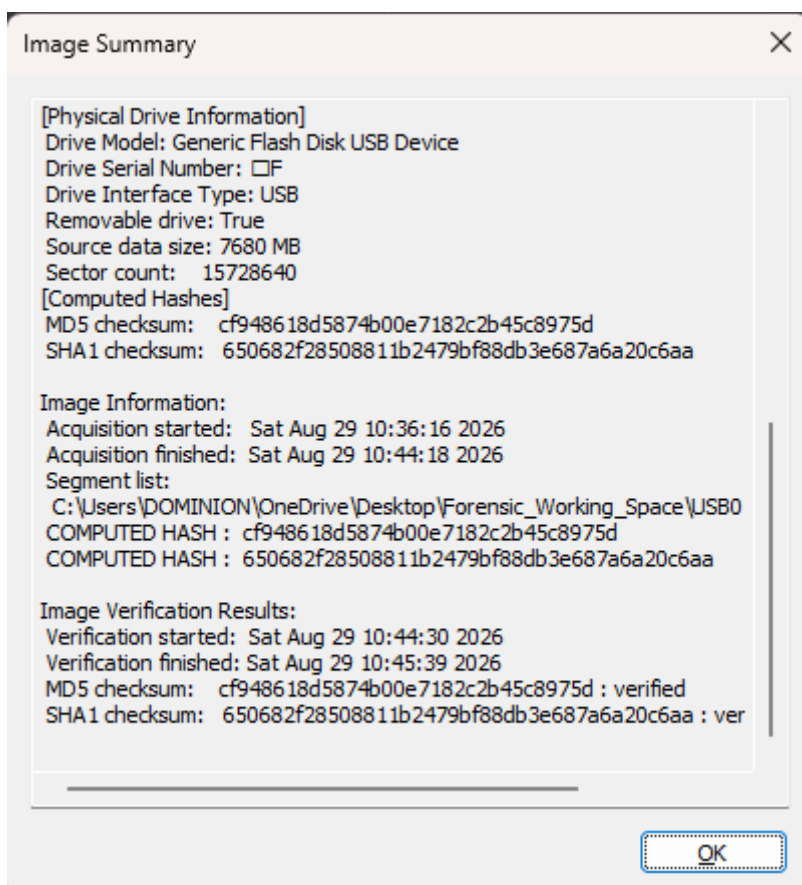


Image Created Successfully



VALIDATING THE TEXT IMAGE AND PASSWORD HASHING

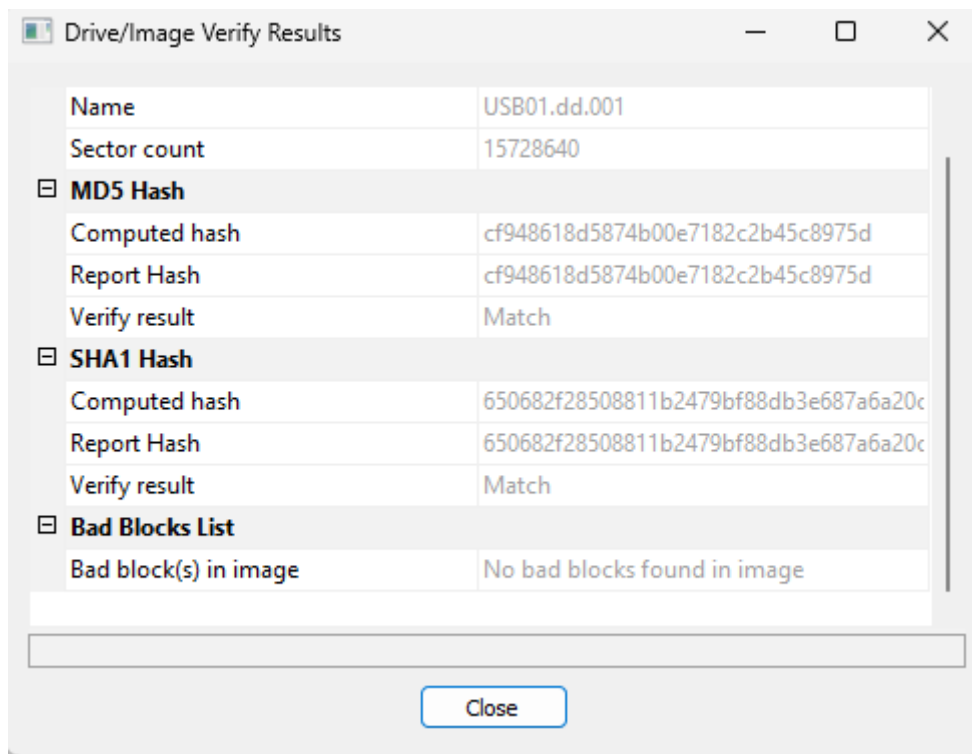


Fig: Password Hashing Validation

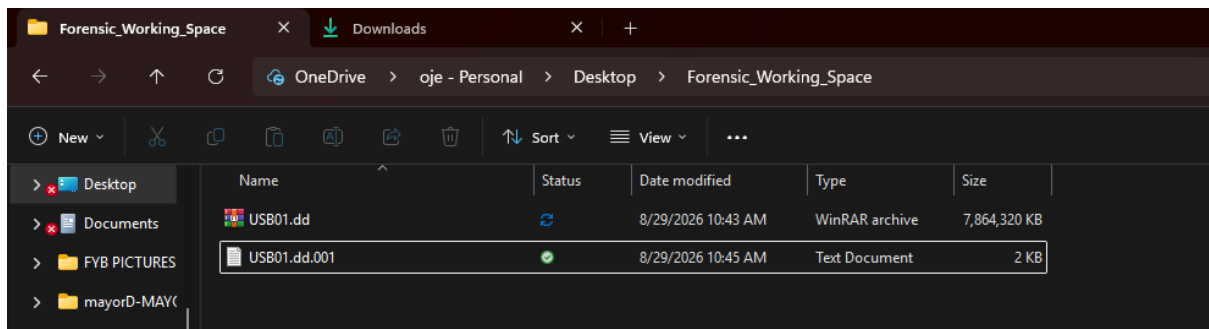


Fig: Verification of Image